BOPLASS

Tauranga LiDAR and RGBI Imagery 2024-2025

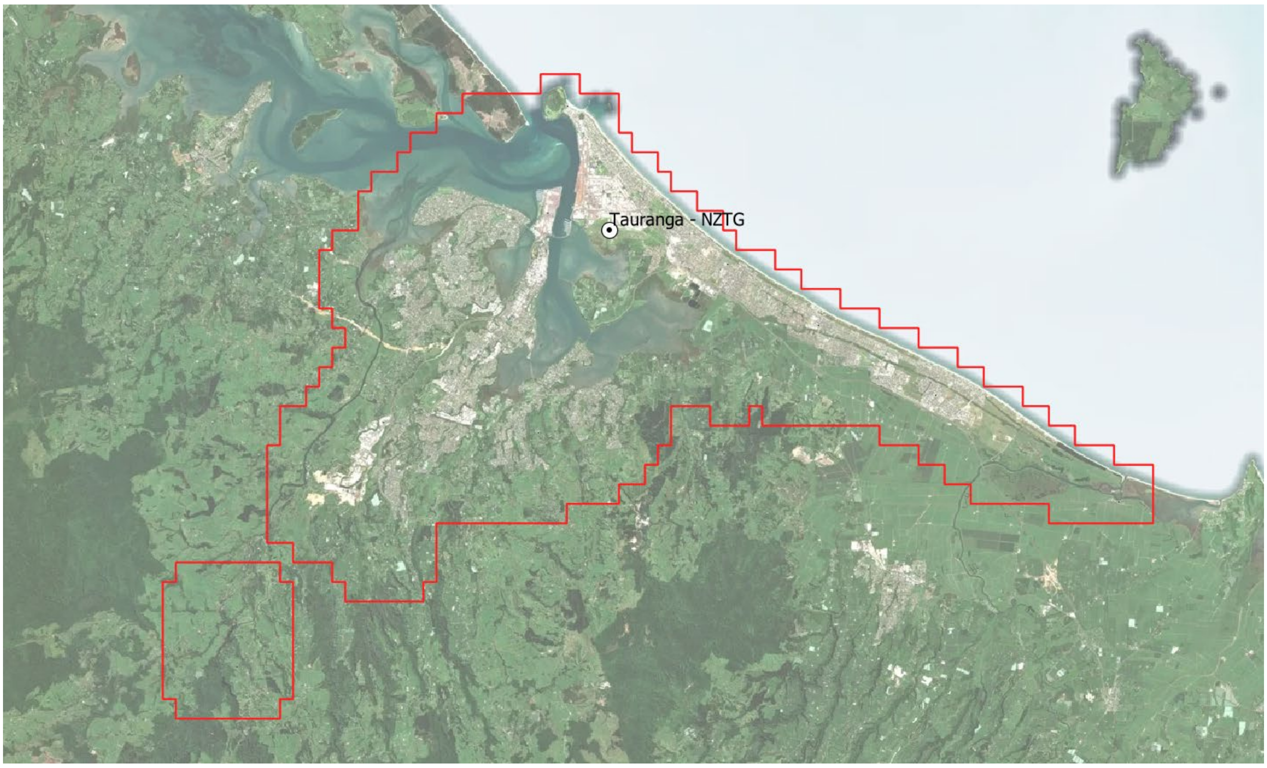
VOLUME: PRJ47237_01-03

PROJECT SUMMARY

Woolpert were engaged by BOPLASS Ltd to undertake acquisition and processing of LiDAR and aerial imagery over Tauranga City, an area of approximately 312 km^2.

This report summarises the information collected, and the products delivered. Acquisition occurred between the 12th of February and 2nd March 2025. (Amendments following LINZ QC have been included.)

This survey was planned to achieve LiDAR data with vertical accuracy +/-10cm RMSE and horizontal +/- 30cm RMS. Emitted pulse density was 8ppsm, and ground classification to ICSM level 2. Concurrent RGBI imagery was collected with 10cm GSD – both imagery and LiDAR were primary products.



Level 1, 6 Ossian Street,
NAPIER 4110
woolpert.com

DATA SUMMARY

This volume includes the following data in NZTM2000 projection, NZVD2016 vertical datum:

- LiDAR Products 902 x NZTopo50 1:1000 tiles
    - Classified Point Cloud data in LAS v1.4 format
    - Ground Point Cloud data in LAS v1.4 format
    - Non-ground Point Cloud data in LAS v1.4 format
    - 1m cell DEM in GeoTIFF format
    - 1m cell DSM in GeoTIFF format
    - Ancillary files in ESRI Shapefile format – Tile Index, Extent, Flightlines, Hydro Breaklines

- Imagery Products 902 x NZTopo50 1:1000 tiles
    - RGBI 10cm Imagery in GeoTIFF format
    - RGBI 10cm Imagery in ECW format
    - RGB 10cm Imagery in ECW format

- 1 x RGB 10cm ECW mosaic

- File listing in text file format
- Metadata file: This document in PDF format

CONTENTS

1. DATA INFORMATION

Data supply: LINZ AWS
Number of files: 13,521 data files, 3 file lists, 1 metadata file
Data formatted on: 24/06/2025(LiDAR), 19/05/2025 (RGBI Imagery), 25/06/2025 (LiDAR updates)
Metadata Document: This file

Previous Deliveries	Date	Title	Contents
PRJ47237_01	20/03/2025	Tauranga LiDAR and RGBI Imagery 2024-25	Quicklook ortho

File Naming in this Delivery	Contents
e.g.CL2_BE37_2025_1000_2041.LAS	Classified point cloud in LAS v1.4 format
e.g.DTM_BE37_2025_1000_2041.LAS	Ground points in LAS v1.4 format
e.g.NON_BE37_2025_1000_2041.LAS	Non-ground points in LAS v1.4 format
e.g.DEM_BE37_2025_1000_2041.tif	1m grid DEM – GeoTIFF format
e.g.DSM_BE37_2025_1000_2041.tif	1m grid DSM – GeoTIFF format
Ancillary files: PRJ47237_Tauranga_LiDAR_AOI_NTZM PRJ47237_Tauranga_LiDAR_TileIndex_NTZM PRJ47237_Tauranga_LiDAR_Trajectory_NTZM PRJ47237_Tauranga_LiDAR_Hydro_3D_Rivers_NTZM PRJ47237_Tauranga_LiDAR_Hydro_3D_Lakes_NTZM PRJ47237_Tauranga_LiDAR_Hydro_3D_Coast_NTZM PRJ47237_Tauranga_LiDAR_Breakline_Bridges_NTZM PRJ47237_Tauranga_Imagery_TileLayout_NTZM PRJ47237_Tauranga_Imagery_Flightlines_NTZM	ESRI Shapefile format: Extent Tile Index Flight lines Hydro Flattening files Hydro Flattening files Hydro Flattening files Bridge enforcement files Tile Index Flight lines – as per LiDAR with 3 extra lines over the CBD

Imagery: RGBI_BD42_2025_1000_1323.tif/ecw RGB_BD42_2025_1000_1323.ecw Tauranga_10cm_Summer2024-25_NZTM.ecw	 RGBI 1:000 tiles in GeoTIFF/ECW format RGB 1:000 tiles in ECW format RGB Mosaic in ECW format
Readme_PRJ47237_01-03.pdf	Metadata Report
PRJ47237_01_LiDAR_FileListing.txt PRJ47237_02_Imagery_FileListing.txt PRJ47237_03_Imagery_FileListing.txt	Lists of product files delivered in this project

2. METADATA

Source Data	Source	Description	Ref No	Date
LiDAR & Imagery	Woolpert	TerrainMapper2 - 527	FL024559	12/02/2025
		TerrainMapper2 – 527	FL024687	22/02/2025
		TerrainMapper2 – 527	FL024692	23/02/2025
		TerrainMapper2 – 527	FL024720	26/02/2025
		TerrainMapper2 – 527	FL024768	02/03/2025
GPS Base Data	GeoNET /LINZ	GeoNET CORS, LINZ GDB co-ordinates TRNG	PRJ47237	As above
Trajectories	Woolpert	GNSS Tightly Coupled Base Station Solution	PRJ47237	
Control	Sounds Surveying	RTK GNSS	PRJ45792	19-23 Feb 2024

LiDAR Characteristics	Description
Format	LAS 1.4
Emitted Density	8 ppm2
Tile size	480m x 720m (NZ Topo 50, 1:1000 tiles)
ICSM Classification	Level 2. Ground surface improvement
Capture Constraints	Coastline captured within 3 hours of low tide. (not applicable in these areas)

No	Point Class	Description	ICSM	CI %
1	Default	Unclassified	1	95
2	Ground	Ground	2	98
3	Low vegetation	< 2.0 m	1	95
4	Medium vegetation	2.0 – 8.0 m	1	95
5	High vegetation	> 8.0 m	1	95
6	Buildings, structures	Buildings, houses, sheds, silos etc.	1	95
7	Low / high points	Spurious low point returns (unusable)	1	95
9	Water	Any point in water	2	98
17	Bridge	Any bridge or overpass	2	98

18	High Noise	Spurious high point returns (unusable)	1	95
22	Temporal Exclusion	Used for data excluded due to a change in surface observed in the capture period, e.g. water in the tidal areas at different heights or earthworks in progress.	1	95

Ortho Characteristics	Description
Format	GeoTIFF
Ground Sample Distance	10cm
Terrain Model	Concurrent LiDAR DEM
Tile size	480m x 720m (NZ Topo50, 1:1000 tiles)
Sample Type	8-bit Integer
Image Bands	RGBI
Orientation/AT	Aero-triangulation Block Adjustment
Capture Constraints	Within 3 hours of low tide Minimum 40 degrees sun angle

Reference Systems	Horizontal	Vertical
Datum	NZGD2000	NZVD2016
Projection	NZTM2000	N/A
Geoid Model	N/A	NZGeoid2016

Accuracy Specification	Measured Point	Derived Point	Basis of Estimation
Control Points	0.05m		Survey Methodology
LiDAR (Horizontal)	0.30m		Project design
LiDAR (Vertical)	0.10m		Project design
Ortho pixels	0.30m		Project design

Notes On Expected Accuracy

- Values shown represent standard error (68% confidence level or 1 sigma), in metres
- “Derived points” are those interpolated from a terrain model.
- “Measured points” are those observed directly.
- Accuracy estimates for terrain modeling by LiDAR refer to the terrain definition on clear ground.
- LASER strikes have been classified into “ground” and “non-ground”, based upon algorithms tailored for major terrain/vegetation combinations existing in the project area. The definition of the ground may be less accurate in isolated pockets of dissimilar terrain/vegetation combinations.

Limitations of Data

- The definition of the ground under trees may be less accurate.
- Building and Vegetation classes are automatically generated, and may contain other items such as fences, vehicles, poles, powerlines.
- Tidal conditions varied over the capture period, but we have aimed to present as much exposed ground as possible and utilised class 22 to hold tidal water overlapping land in the neighbouring swath.

Workflow Summary

- GNSS Processing: Applanix POSPAC MMS
- Raw Data Processing: HzMap for initial processing and StripAlign for swath matching
- Coverage Review: complete coverage was obtained (note 3 lines over the CBD were for Imagery only)
- Point Density Review: point density met or exceeded specification over 86% of the AOI, areas of lower density were examined and found to be areas of low reflectivity – water or dark composite roofing
- Control Analysis – the dataset was compared to both vertical and horizontal control in the area. Results are presented in the next section.
- Classification: classification accuracy was verified by comparative analysis of 15 of 100m x 100m randomised sample sites within the project AOI. All ground areas were edited by an independent party to accuracies exceeding 98%. All non-ground features were edited to achieve 95% accuracy by the project’s LiDAR analyst, using automated methods.
- Point Cloud and Raster Statistics: All LiDAR products have been checked against the project specifications and have passed Woolpert’s internal QC
- The LiDAR DEM has been used for imagery rectification, seamlines between images have been generated automatically and reviewed manually to obtain the desired output. Three additional runs were collected over Tauranga CBD and Mount Maunganui hotels to limit building displacement in these areas.

Data Validation – LiDAR Data

- Vertical Accuracy Validation - Ground data has been compared to ~270 test points obtained by field survey and assumed to be error-free. The test points were distributed in 6 groups across the mapping area and located on clear ground. Comparison of the test points with elevations interpolated from measured data resulted in:

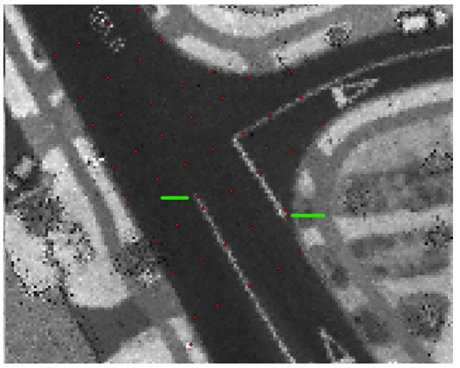
Site	dZ	RMS	SD	Calc	Post dZ	Post RMS	Post SD
Average	*0.150*			*0.000*	*0.00*		
Control_Site1	0.177	0.178	0.022	0.027	0.027	0.034	0.022
Control_Site2	0.179	0.181	0.028	0.029	0.029	0.040	0.028
Control_Site3	0.153	0.155	0.021	0.003	0.003	0.021	0.021
Control_Site4	0.118	0.120	0.023	-0.032	-0.032	0.039	0.023
Control_Site5	0.133	0.135	0.021	-0.017	-0.017	0.027	0.021
Control_Site6	0.141	0.143	0.027	-0.009	-0.009	0.028	0.027

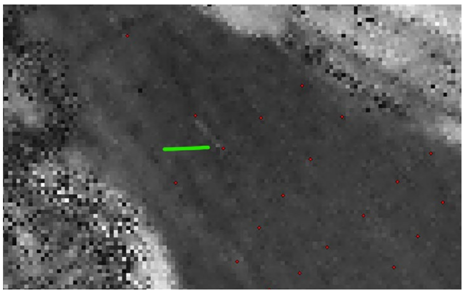
The mean elevation difference of -0.150m has been applied to the data supplied in this volume.

Comparison of all points with elevations interpolated from measured data resulted in:

Mean difference: **0.000m**
St. Deviation: **0.033**
Standard Error (RMS): **0.033m or 0.064m @ 95% CI**

- Horizontal Accuracy – the LiDAR data was compared to visible horizontal control using the intensity imagery. The data was found to fit well. The expected horizontal accuracy is 20cm at 1 sigma.





- Data classification has been manually checked and edited against available imagery.

Data Validation – Orthophoto Image

- This data has not been field tested for accuracy. Comparison of the orthophoto with control points which were also used in the photo orientation process resulted in a horizontal standard error (RMSE) well within 30cm. Full proof of accuracy achieved requires comparison to independent test points.

Required Accuracy (RMS)	0.300m
RMS_X	0.047m
RMS_Y	0.071m
RMS Positional XY	0.085m
Positional Accuracy 95% CI	**0.167m**
Number of Points	31
Type of GCP	Control Point/existing features

3. CONDITIONS OF SUPPLY

The data in this volume has been commissioned by the **BOPLASS Ltd**.

The data in this volume is provided by Woolpert to **BOPLASS Ltd** under the Terms of Engagement described in **MQM022 Terms of Engagement NZ Creative Commons**. Which provides the client, **BOPLASS Ltd** with full and unrestricted license in perpetuity to use all reports, data and delivered data for the purpose set out in the definition of services. The Client is also permitted to freely grant nonexclusive, royalty-free sub-licenses to third parties in respect of the delivered data pursuant to the Creative Commons Agreement by Attribution 4.0 New Zealand. The Client must ensure that **Woolpert NZ Limited** is acknowledged as the copyright owner in the delivered data.

1. This file (Readme_PRJ47237_01-03.pdf) will always be stored with the unaltered data contained in this volume.

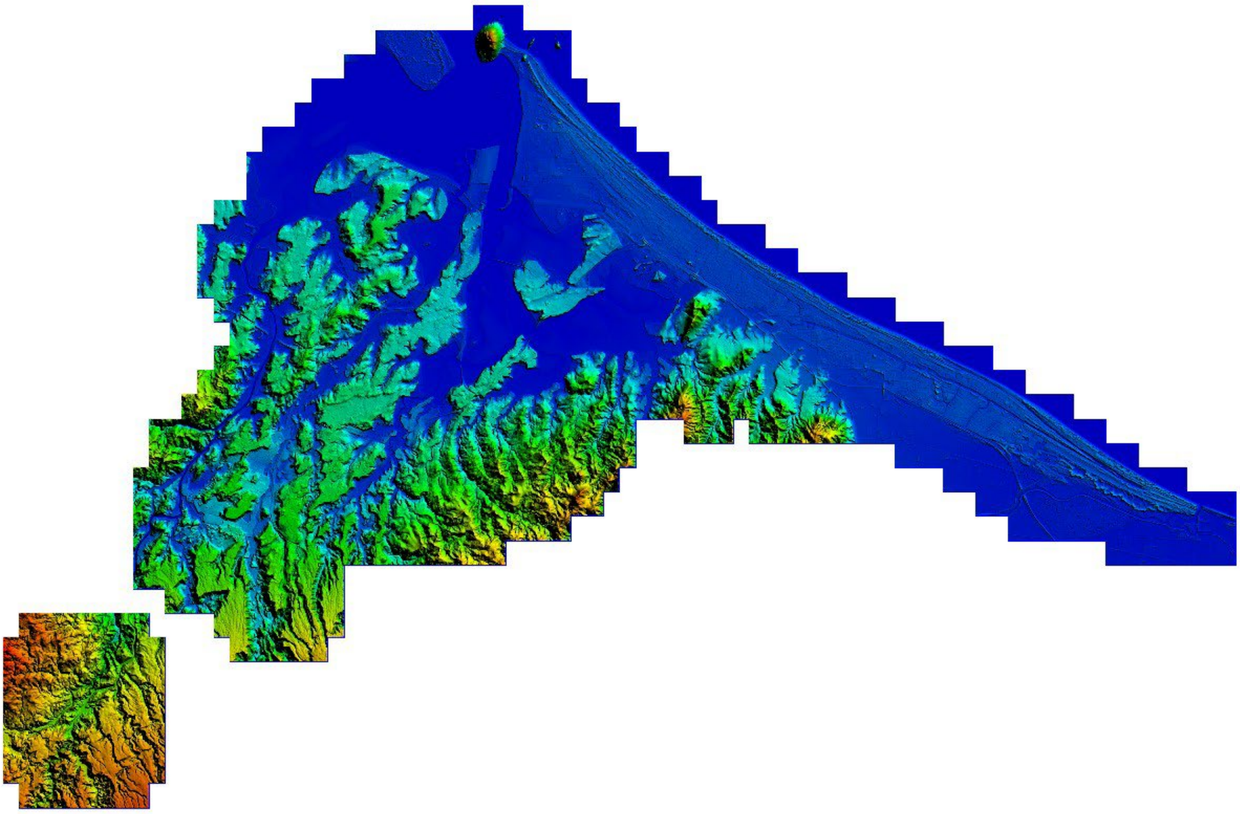
This data is provided in accordance with the specifications agreed with BOPLASS Ltd and Tauranga City Council. Any problems associated with the information in the data files contained in this volume should be reported to Woolpert, Asia-Pacific.

Woolpert NZ Limited
Level 1
6 Ossian St
Napier 4110
New Zealand

woolpert.com
Twitter Facebook LinkedIn Instagram

4. VALIDATION

Tauranga LiDAR – DEM Colour Elevation Image



Tauranga Imagery